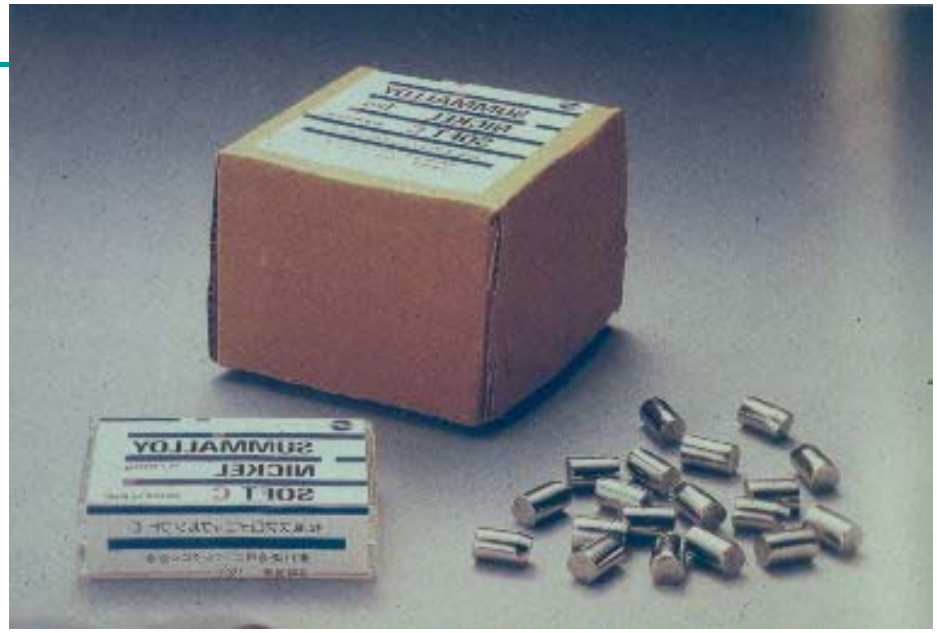
METALS

Alloy: a mixture of two or more metals

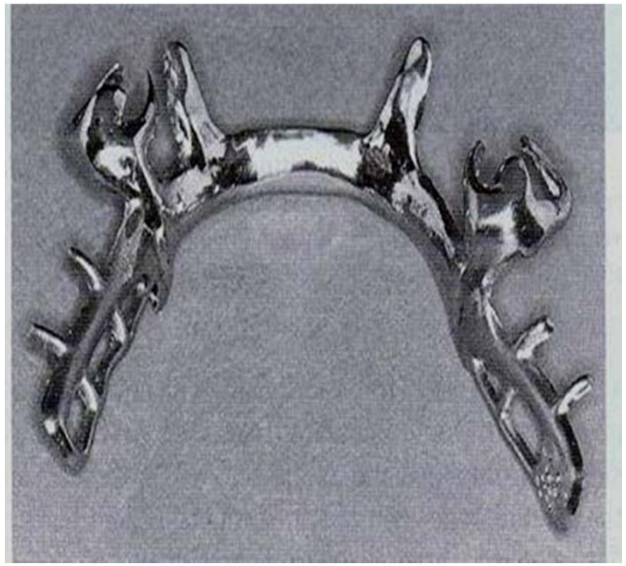
Pure metals are rarely used in dentistry because they are <u>weaker</u> than they are when mixed with other metals.

Metals in dentistry
Applications of metal alloys

Dental material



Cast metal restorations



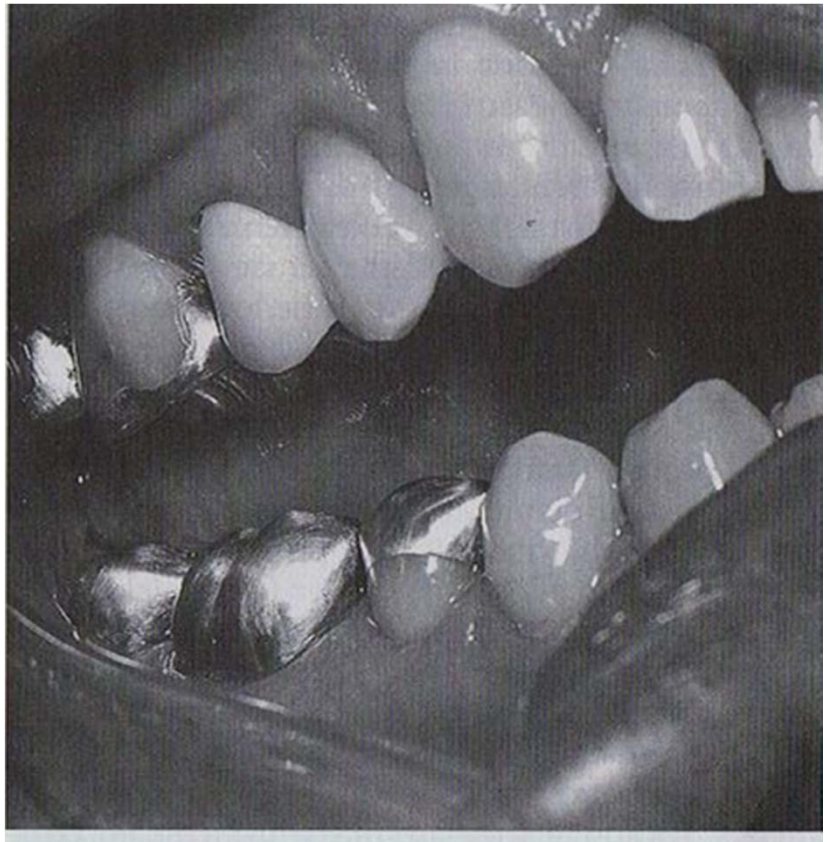
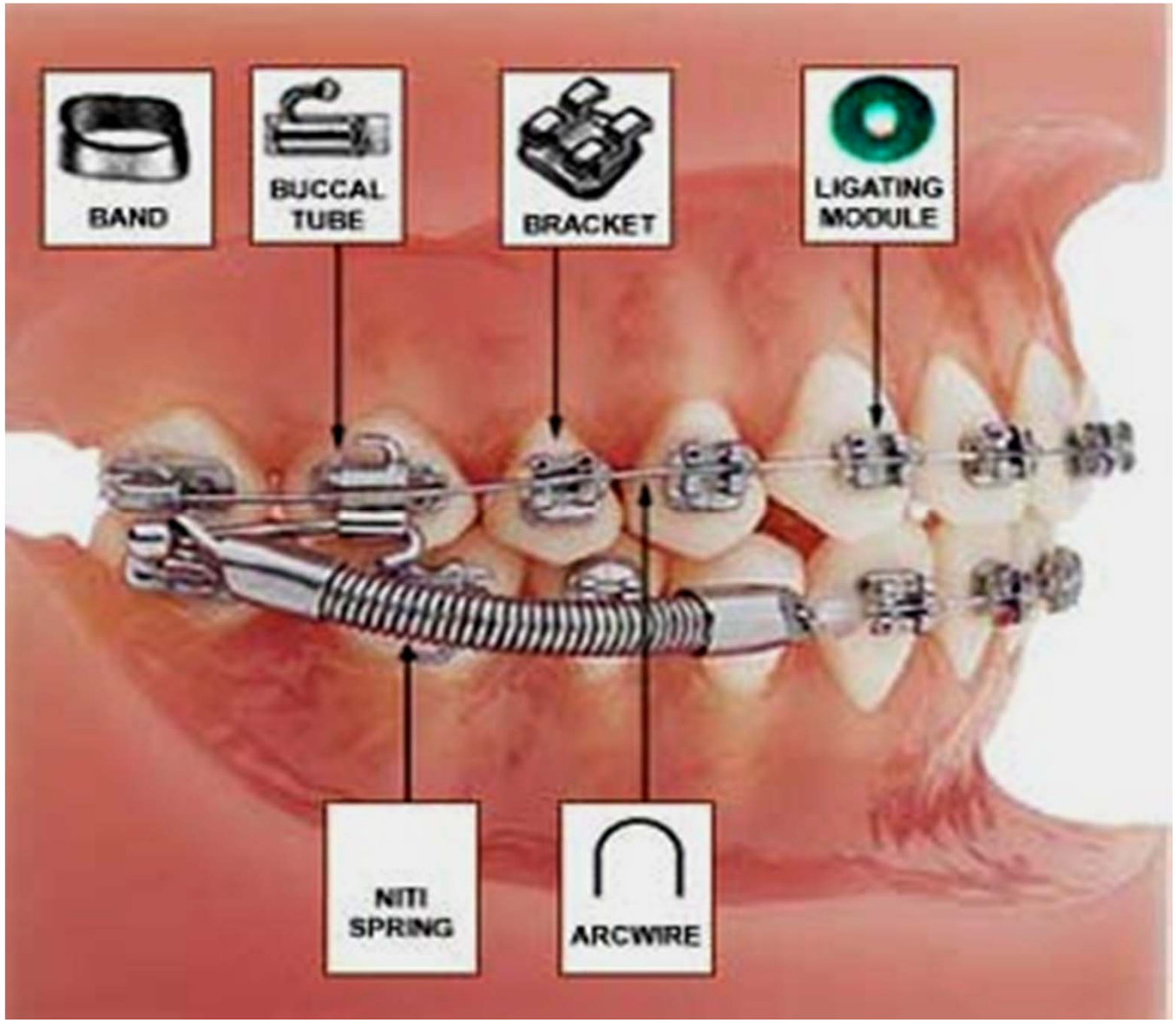
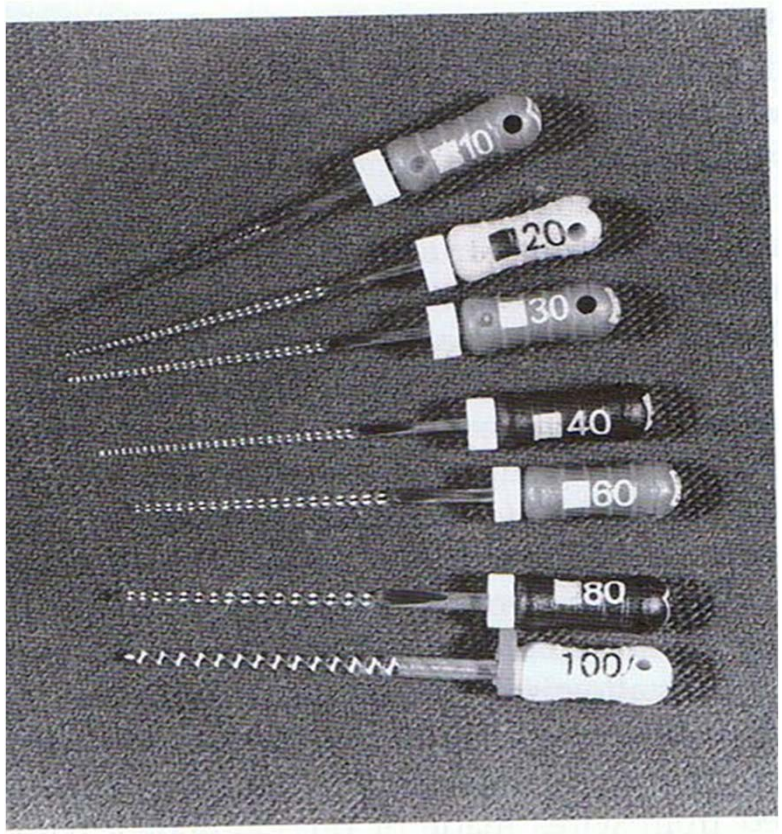
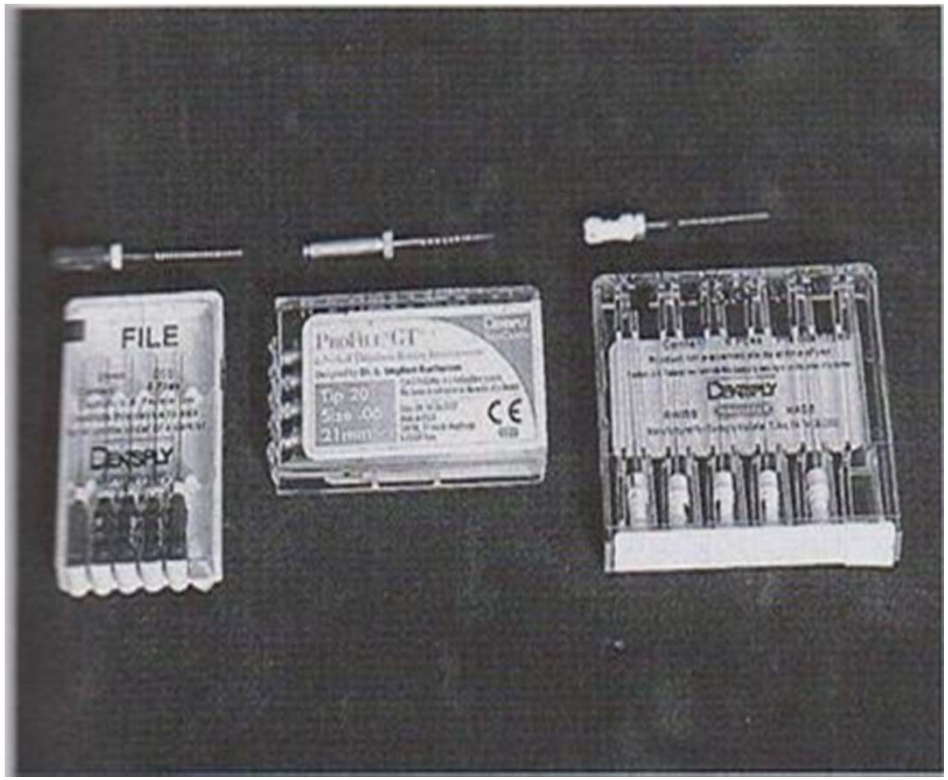


FIGURE 8-4 ■ Partial- and full-coverage cast gold restorations


BAND
BUCCAL TUBE
BRACKET
LIGATING MODULE
NITI SPRING
ARCWIRE

Endodontic files





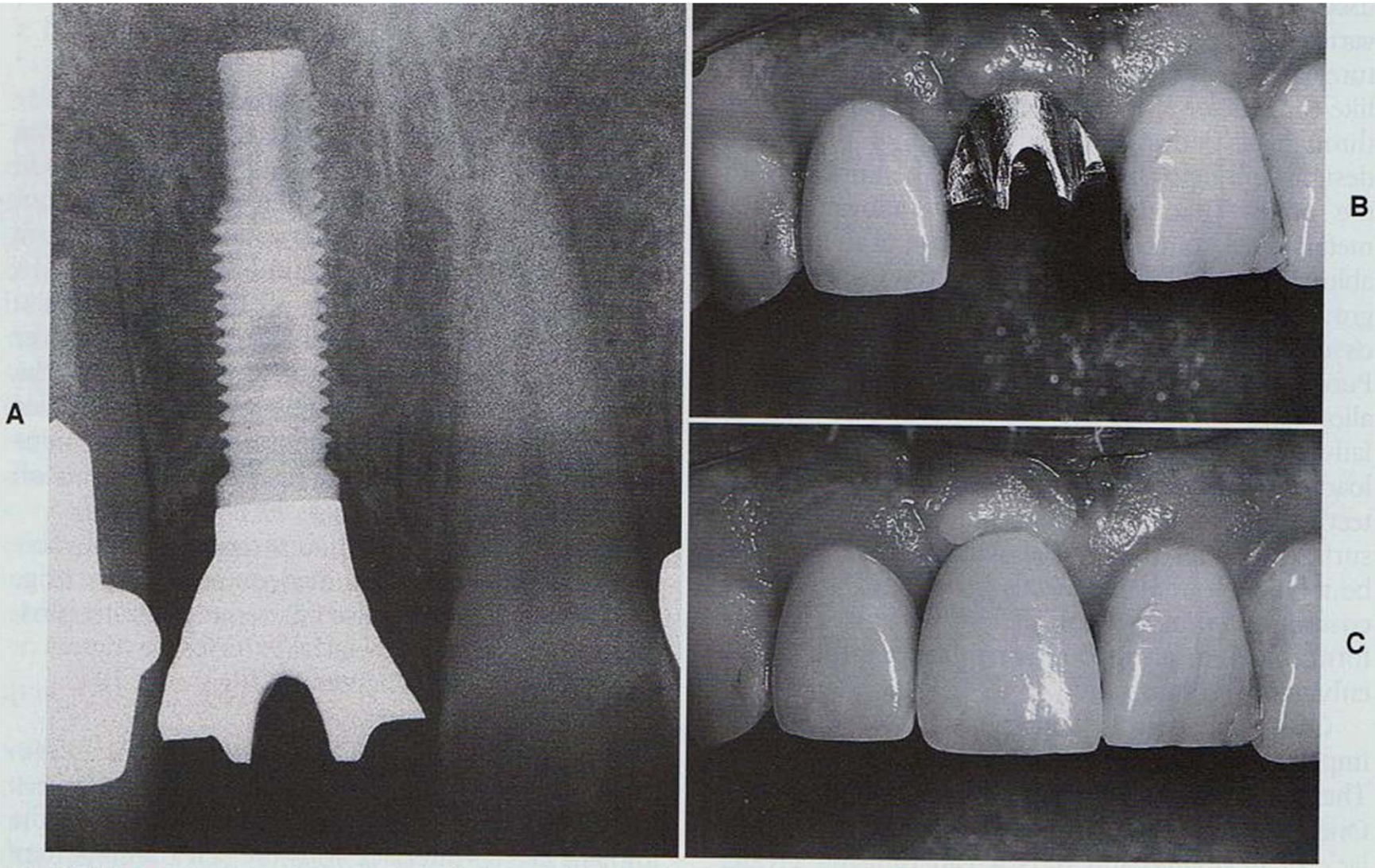


FIGURE 8-8 ■ Single-tooth implant. A, Radiograph of implant fixture in bone. B, Abutment member to which the crown is attached. C, Crown attached to implant.

Requirement of metals alloy

1-Metals must not tarnish or corrodes in the mouth.

2- Metals must be sufficiently strong

3. biocompatible: not toxic and not irritant or allergic.

4- easily to fabricate: melt, cast, cut and grind.

5. Ability to flow:

6. High hardness, high strength and high melting temperature due to metallic bonding

8. They are malleable and Ductile

9. Give a dimensional stability:

10. Easily to solder: Two metals are united together during repair.

Classification of metals

1. noble metal

2. Non noble metal.

Classification of alloys

A// According to the number of alloying elements

1. Binary (2-constituents)
2. ternary(3-constituents)
3. quaternary (4-,constituents) … etc.

B// according to the nobility

1. **High noble alloys 40 %gold or 60 % noble**
2. **Noble alloys 25% noble metal**
3. **Base metal alloys less 25% noble**

C//Classification of dental casting alloy according to the major elements

- Gold
- Silver
- Palladium
- Nickel
- cobalt
- Titanium

D//Classification of dental casting alloy according to Three major elements

- Nickel-Chromium-molybdenum
- Gold-palladium-silver
- palladium-silver-tin
- Nickel-Chromium-molybdenum
- Iron-Nickel-Chromium
- Titanium-Aluminum-vanadium

Alloys in Dentistry

1. Gold alloys (solid solutions) are used for inlays, crowns and bridges. Alternatives to gold alloys include silver palladium, and nickel-chromium alloys.
2. Cobalt chromium alloys are used for partial denture bases.
3. Silver –copper alloys (eutectic alloy) is used in dental soldering.
4. Nickel alloy.
5. Titanium alloy.
6. Palladium alloy.

Gold (Au)

1. Pure gold
2. Gold alloy

Most corrosion resistant

Pure gold is 24 karat, 100%, or 1000 fine (percentage * 10)

Pure gold is too soft, so, gold alloys were mostly used

CLASSIFICATION OF gold ALLOYS

a. ADA Classification System
   (1) Type I soft small inlay
   (2) Type II medium large inlay, onlay.
   (3) Type III -- onlay, crown
   (4) Type IV extra hard -- crown, bridge, removable partial dentures

b. Effects of Alloys Components:

(1)	Gold	(Au) →	Corrosion resistance
(2)	Copper	(Cu) →	Hardness, red color.
(3)	Silver	(Ag) →	Counteract orange color of copper
(4)	Palladium	(Pd) →	Increase MP and hardness
(5)	Platinum	(Pt) →	Increase MP
(6)	Zinc	(Zn) →	Prevent oxidation during melting (O_2 getter)

properties

- Color : yellow or white depending on ……
- corrosion resistance better because of higher noble metal content.
- Ductility and malleability
- strength.
- Hardness : type III (121) Mpa and type IV (149) Mpa
- Investment: gypsum bonded investment.
- Density of the pure gold (19.3) gm/cm and alloys have less density.
- percent elongation type III (30-40%) and type IV (30-40%) Mpa.
- Modulus of elasticity (150-210) Mpa so casting can made lighter and thinner.
- Strength type III (207) Mpa and type IV (275) Mpa
- Bonding to porcelain : formed good oxide layer that has good bonding to porcelain
- Contraction shrinkage less than 1.25%-1.65%.
- biocompatibility : relatively biocompatible.
- Fusion temperature : (920-960 C)

Alternative to gold alloys

- A. silver- palladium alloys cheaper white in color
- B. titanium and titanium alloys

Metal ceramic alloys

- They are metal computable with ceramic and bond to it.

- Types of metal ceramic alloys
- High noble (gold alloy) : gold-palladium, gold-palladium-platinum, gold-palladium-silver.

- Noble (palladium) : palladium-silver, palladium-gold, palladium-copper.

- Base metal: nickel chromium, pure titanium, titanium-aluminium-vanadium.

Removable denture alloys

- requirement
  - Low weight
  - High stiff
  - High fatigue resistance
  - Low cost
  - Not reacted with denture cleanser

Types of RPD

- Cobalt chromium
- Nickel chromium
- Titanium alloys
- Aluminum alloys
- Type IV gold alloys

Removable prosthetic casting alloys

Base metals used (Cobalt Chromium Alloy)

Cobalt (35- 65%) decrease hardness, strength and rigidity.

Chromium (23-30%) passive effect and decrease melting point.

Nickel (0-20%) decrease hardness and strength and increase ductility and cause sensitivity.

Iron (0-5%) increase hardness

Carbon (0.4%)

Molybdenum (0-7%) increase hardness

These characteristics include:

- Density half of the gold (8-9gm/cm)
- Fusion temperature higher than gold (1250-1480 C^o)
- Strength stronger than gold 710 Mpa
- percent elongation less than gold (1-12%)
- Modulus of elasticity twice than gold (225 Mpa)
- Hardness harder than gold (432 HN)
- corrosion resistance better (passive layer formation chromium oxide)
- Contraction shrinkage of about 2.3%.

Removable prosthetic casting alloys

Base metals used (Nickel Chromium Alloy)

	Iron (0-5%)
Copper (minor)	Beryllium (minor)
*Chromium (11-27%)	Sillicate
*Nickel (61-81%)	Carbon (0.4%)
Aluminum (small)	Molybdenum (2-9%)

These characteristics include:

- Color white in color.
- Density half of the gold (7.8-8.4) gm/cm
- Melting range higher than gold (1155-1304 C^{o})
- Hardness harder than gold (175-360 VHN) so difficult to cut, grind and polish
- Strength stronger than gold (310-828) Mpa.
- Modulus of elasticity (150-210) Mpa so casting can made lighter and thinner.
- percent elongation less than gold (10-28%) ductile but not easily burnishable.
- Bonding to porcelain : formed good oxide layer that has good bonding to porcelain
- Aesthetic : the dark oxide layer may show at the metal porcelain junction.

Titanium and titanium alloy

- Color white in color.
- corrosion resistance better because of oxide layer and passivity.
- Strength : good
- Weight: light
- Investment: phosphate and ethyl silicate bonded investment is used.
- Density : light in weight (1-4) gm/cm
- Modulus of elasticity (110) Gpa so half rigid than base metal alloys.
- Bonding to porcelain : formed good oxide layer that has good bonding to porcelain
- Coefficient of thermal expansion 8.3cm/cm^{c} small compared to porcelain 12.7-14.2 so need special low fusing porcelain to work with it.
- Biocompatibility: not toxic and excellent to contact with hard and soft tissues
- Fusion temperature higher than gold (1668)C so need special equipment.

Wrought metals

An alloy which are shaped without applying heat (room temp.) by hammering cold working

Thank You
A Dreamy Wor
A man's dreams are an index to his great